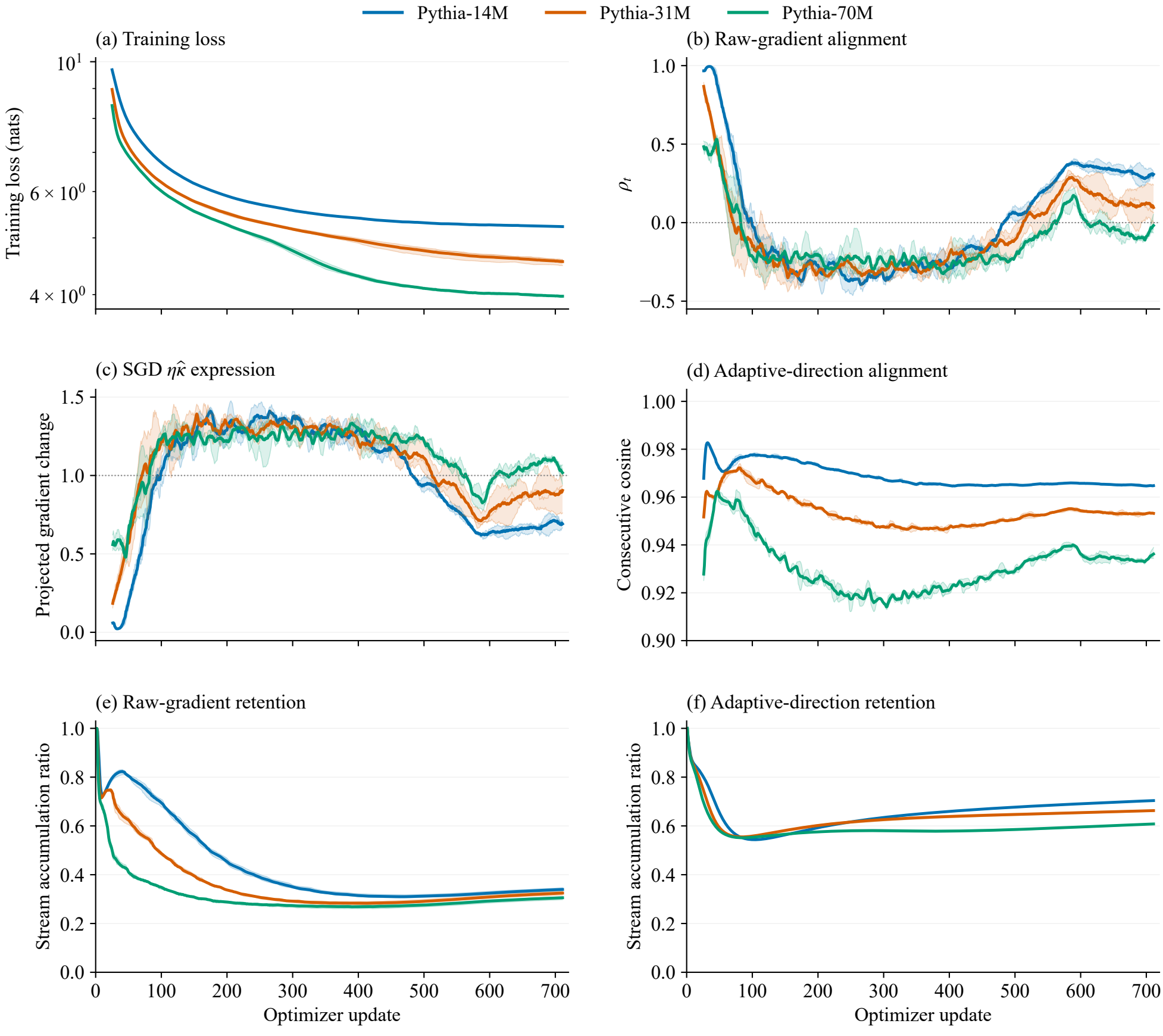

Pythia-14M
Pythia-31M
Pythia-70M
(a) Training loss
Training loss (nats)
10^1
6×10^0
4×10^0
(b) Raw-gradient alignment
ρ_t
1.0
0.5
0.0
−0.5
(c) SGD $\eta\hat{\kappa}$ expression
Projected gradient change
1.5
1.0
0.5
0.0
(d) Adaptive-direction alignment
Consecutive cosine
1.00
0.98
0.96
0.94
0.92
0.90
(e) Raw-gradient retention
Stream accumulation ratio
1.0
0.8
0.6
0.4
0.2
0.0
0
100
200
300
400
500
600
700
Optimizer update
(f) Adaptive-direction retention
Stream accumulation ratio
1.0
0.8
0.6
0.4
0.2
0.0
0
100
200
300
400
500
600
700
Optimizer update